ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3, Q4 (Q3 struck out – not to be considered)

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 μm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) (i) $f_x = f/p = 16\times10^{-3} / 5\times10^{-6} =$ **3200 px**.

(ii) Sensor width = 1280 × 5 μm = 6.4 mm. Using similar triangles, FOV = $W_s \cdot f / Z = 6.4 \times 16/400 =$ **0.256 mm**.

Distance per pixel = FOV/1280 = 0.256/1280 = $\mathbf{2\times10^{-4}}$ **mm** = 0.2 μm.

(b) Resolutoin of the ADC = $V_{range}/2^n = 5$ V / 4096 = 1.22×10^{-3} V = **1.22 mV**.

So the smallest change the system can resolve is 1.22 mV.

(c) Using forward Euler: $x_{k+1} = x_k + T(A\,x_k + B\,u_k)$, so F = 1 + AT and G = BT.

F = 1 + (−2)(0.1) = **0.8**, G = 1 × 0.1 = **0.1**.

$x_{k+1} = 0.8\,x_k + 0.1\,u_k$.

I am not very confident about this one, will see if time permits to revisit.

Question 2

Given: J = 0.02 kg·m², b = 0.05 N·m·s/rad, m = 1 kg, ℓ = 0.25 m, g = 9.81 m/s². So mgℓ = 1 × 9.81 × 0.25 = 2.4525 N·m (used throughout).

(a) The model is second order, so I choose the position and velocity as states: $x = [x_1, x_2]^T = [\theta, \dot{\theta}]^T$, with input τ.

Rewriting $J\ddot{\theta} + b\dot{\theta} + mg\ell \sin\theta = \tau$ gives the nonlinear state-space model: $\dot{x}_1 = x_2$, $\dot{x}_2 = (1/J)(\tau - b\,x_2 - mg\ell \sin x_1)$, i.e. $\dot{x} = f(x, \tau)$.

Equilibria for constant τ_0 need $\dot{x} = 0$: the first equation gives $x_2 = 0$, and the second then gives $mg\ell \sin\theta_e = \tau_0$, i.e. $\sin\theta_e = \tau_0/(mg\ell)$.

This has a solution only if $|\tau_0/(mg\ell)| \le 1$, so $\mathbf{|\tau_0| \le mg\ell = 2.4525\ N\cdot m}$. For $|\tau_0| < mg\ell$ there are two equilibria per revolution (θ_e and $\pi - \theta_e$); at $|\tau_0| = mg\ell$ they merge, and beyond it the joint cannot be held statically.

(b) $A = [0, 1; -(mg\ell/J)\cos 60°, -b/J] =$ **[0, 1; −61.31, −2.5]**.

$B = \partial f/\partial\tau = [0; 1]$ (the torque enters directly) so **B = [0; 1]**.

(c) Set all time derivatives to zero in the closed loop. With $\tau = K_p(\theta_d - \theta) - K_d\dot{\theta}$ and $\dot{\theta} = \ddot{\theta} = 0$ the damper term vanishes (so K_d has no effect on the final position) and the dynamics reduce to $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss})$.

Why $\theta_{ss} \neq \theta_d$: if the joint were at rest exactly at $\theta_d = 90°$ the PD torque would be $K_p \cdot 0 - K_d \cdot 0 = 0$, yet graviy pulls with $mg\ell \sin 90° = 2.4525$ N·m. The net torqe would be non-zero, so the joint could not be at rest. A purely proportional-derivative law can only hold a load if there is a stedy error.

Let $e = \theta_d - \theta_{ss}$ (in rad). Then $\sin\theta_{ss} = \sin(\pi/2 - e) = \cos e$, and the equation becomes $50\,e = 2.4525 \cos e$. Fixed-point iteration: $e_0 = 2.4525/50 = 0.04905$ (taking $\cos e \approx 1$); $e_1 = 0.04905 \times \cos(0.04905) = 0.04899$; $e_2 = 0.04905 \times \cos(0.04899) = 0.04899$, converged.

e = 0.04899 rad × (180/π) = 2.807°, so $\theta_{ss} = 87.19°$. **Steady-state error ≈ 2.81° (the arm sags below the target).** This is small because K_p is much larger than $mg\ell$ ($K_p/mg\ell \approx 20$).

(d) New steady-state error = **0**, because the added term cancels the graviy torque exactly.

The poles do not change, because the added term is a feed-forward term that does not depend on the error, so it cannot affect the stability of the loop.

Question 3 [CROSSED OUT BY STUDENT]

~~*Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is T = 1/f_s = 1 ms.*~~

~~**(a)** (1) Diferentiation amplifies noise in the encoder signal.~~

~~(2) Finite differencing needs a lot of memory and processing, which is slow on a micro-controller at 1 kHz.~~

~~**(b)** ω_c = 20 rad/s. Tustin with T = 0.001: a = ω_cT/2 = 0.01.~~

~~$y_k = [a/(1 + a)](u_k + u_{k-1}) + [(1 − a)/(1 + a)]\, y_{k-1}$ = **0.00990 (u_k + u_{k-1}) + 0.98020 y_{k-1}**.~~
~~DC gain: 2(0.0099)/(1 − 0.9802) = 0.0198/0.0198 = 1 ✓.~~

~~**(c)** A − LC = [$-\ell_1$, 1; $-\ell_2$, −2]; characteristic polynomial det(sI − A + LC) = (s + ℓ_1)(s + 2) + ℓ_2 = s^2 + (2 + ℓ_1)s + (2ℓ_1 + ℓ_2).~~
~~Desired $(s + 10)^2 = s^2 + 20s + 100$. Then 2 + ℓ_1 = 20 ⇒ ℓ_1 = 18, and 2ℓ_1 + ℓ_2 = 100 ⇒ ℓ_2 = 100 − 18 = 82.~~

~~**L = [18, 82]^T**.~~

~~**(d)** Advantage of the observer: it uses the model and the input u, so there is no lag in the estimate, which a filter always introduces.~~

~~The filter is safer when the sampling rate is very high, becuase the observer would not be able to run fast enough.~~

~~*(This question is cancelled, please do not evaluate it.)*~~

Question 4

Given: plant $\ddot{\theta} + a\dot{\theta} + c\theta = b\,u$ with a = 2, c = 6, b = 1; controller $u = N_{ff}\, r - k_1\theta - k_2\dot{\theta}$.

(a) Substitute the control law into the plant: $\ddot{\theta} + a\dot{\theta} + c\theta = b\,N_{ff}\,r - b\,k_1\,\theta - b\,k_2\,\dot{\theta}$. Moving the feedback terms to the left: $\ddot{\theta} + (a + b\,k_2)\,\dot{\theta} + (c + b\,k_1)\,\theta = b\,N_{ff}\,r$.

Numerically, $a + b\,k_2 = 2 + 2 = 4$ and $c + b\,k_1 = 6 + 2 = 8$, so the characterstic equation is $s^2 + 4s + 8 = 0$. Roots: $s = [-4 \pm \sqrt{(16 - 32)}]/2 = -2 \pm 2j$. They have negative real parts, so the loop is stable; $\omega_n = \sqrt{8} = 2.83$ rad/s and $\zeta = 4/(2 \times 2.83) = 0.707$.

Transfer function from r to θ: $T(s) = b\,N_{ff} / (s^2 + 4s + 8)$. DC gain (s = 0): $T(0) = N_{ff}/8$. For unitiy DC gain, **N_{ff} = 8**. (Note that this is larger than 1 because the graviy-related stiffness c and the feedback k_1 both resist the displacement.)

(b) $G_{DC} = bN_{ff}/(c + bk_1) = 6/7.5 = 0.8$, so θ_{ss} = **0.8**.

Percentage error = 0.8 × 100 = **80%**.

(c) Take the state $z = [\theta, \dot{\theta}, \xi]^T$. The plant gives $\ddot{\theta} = -a\dot{\theta} - c\theta + b\,u$ with $u = -k_1\theta - k_2\dot{\theta} + K_I\xi$, and $\dot{\xi} = r - \theta$. With r = 0 (r only shifts the equilibrium) the closed-loop matrix is A_{cl} = [0, 1, 0; −(c + bk_1), −(a + bk_2), bK_I; −1, 0, 0].

Characterstic polynomial $\det(sI - A_{cl})$, expanding along the first row: $s\cdot[(s + a + bk_2)s - 0] + 1\cdot[(c + bk_1)s + bK_I] = s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$. **(Result)**

Desired pols −4 and −2 ± 2j: $(s + 2 - 2j)(s + 2 + 2j) = (s + 2)^2 + 4 = s^2 + 4s + 8$. Multiply by (s + 4): $s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 24s + 32$.

Compare coefficients with a = 2, b = 1, c = 6: s^2: 2 + k_2 = 8 ⇒ **k_2 = 6**. s^1: 6 + k_1 = 24 ⇒ **k_1 = 18**. s^0: bK_I = 32 ⇒ **K_I = 32**.

Check: all coefficients of the desiered polynomial are positive and (8)(24) = 192 > 32, so Routh–Hurwitz confirms the three poles are in the left half plane ✓.

(d) Becuase the closed loop is stable, the stedy-state error is zero for any b and c.

Failure mode: integrator windup. Remedy: clamp the integrator output (anti-windup).